Injury-to-Regeneration Coupling in Xenopus laevis: A Testable Hypothesis for the Refractory Period

SIGMA AI-assisted research synthesis

Abstract

Why Xenopus laevis tadpoles temporarily lose tail-regeneration competence requires an explanation that distinguishes developmental stage from the regenerative behavior of individual animals. A candidate hypothesis is that the refractory period weakens the coupling between early wound responses and subsequent activation of a transient epidermal regeneration program, rather than simply reducing injury signaling. This hypothesis predicts weaker later regeneration-initiating-cell and extracellular-matrix activation at comparable early wound-response intensity in refractory animals.

However, the available evidence does not establish this conditional relationship, its novelty, a numerical effect, or predictive improvement over known-mechanism baselines. It also provides no independent biological confirmation. The proposed article therefore develops a bounded hypothesis and its falsification requirements, not a newly discovered mechanism. Its contribution is to distinguish a potentially informative temporal dependency from stage classification, tissue-composition effects, and rediscovery of known regenerative programs.

Evidence coverage limitation

The evidence index omitted 1863 material saved research field fragment(s) across 6 source(s). These fragments were not reviewed for this paper. Interpretations that could depend on omitted material remain provisional; the complete frozen source snapshot and omission ledger are retained with the generation record.

1 Introduction and biological context

The motivating question concerns the loss and later recovery of tail-regeneration competence in Xenopus laevis, including variation between animals. A useful explanation must distinguish a developmental change in the relationship between injury and regeneration from a classifier that merely recognizes developmental stage. Explaining temporary failure also requires accounting for subsequent recovery, rather than treating competence as an irreversible developmental decline.

The project brief identifies transient regeneration-initiating cells, abbreviated RICs, their basal-epidermal origin, and extracellular-matrix remodeling as prior findings rather than novel discoveries. The proposed contribution is therefore a conditional relationship: whether a comparable early wound response produces different later epidermal regenerative activation at competent and

refractory stages. These background assertions remain project-supplied context here, because the supplied selection does not contain the underlying publications or a completed prior-art assessment.

Four explanations remain in contention: failure to induce the epidermal regenerative transition, failure to couple injury signaling to that transition, metabolic or tissue-composition gating, and analytical or developmental confounding. The present evidence cannot rank them empirically. The article should consequently take the form of a hypothesis-focused analysis with explicit identification and validation requirements, not a report of a successful transcriptomic discovery.

2 Methods and evidence

The synthesis distinguished biological context, proposed analyses, and records of completed research in the supplied evidence index. Records stating that no complete research artifact or objective result was established were treated as limits on inference, not as negative biological experiments. Strategy descriptions supplied the scope of the proposed comparisons but were not treated as evidence that data retrieval, statistical analysis, or perturbation had occurred.

The proposed temporal-coupling argument was assessed against its stated requirements for sample verification, biological replication, baseline comparison, alternative explanations, novelty assessment, and independent validation. The supplied selection contains no verified sample manifest, expression results, fitted models, quantitative uncertainty estimates, or experimental outcomes. Accordingly, the synthesis identifies what remains untested without assigning numerical effects or claiming accession-level coverage.

3 A conditional hypothesis for the developmental switch

The developmental switch in Xenopus laevis tail regeneration can be framed as a possible failure to translate an early injury response into a later regenerative epidermal state, rather than simply a loss of injury responsiveness. The candidate hypothesis is directional: at comparable early wound-response intensity, refractory-stage wound epidermis would activate a weaker subsequent regenerative program than competent-stage epidermis. This formulation separates the magnitude of the initial response from its relationship to a later transition, while retaining the distinction between developmental stage and the regeneration outcome of an individual animal.

The relevant downstream state is already part of the reported biology, not the proposed discovery. A 2024 Genome Biology study identified transient regeneration-initiating cells (RICs) originating from basal epidermal cells and reported extracellular-matrix remodeling defects associated with regeneration failure (doi:10.1186/s13059-024-03396-3). Extracellular matrix (ECM) denotes the material outside cells that forms their surrounding tissue scaffold; here, RIC/ECM activation refers to the transient epidermal program and associated matrix-remodeling activity. Identifying these cells or finding reduced expression of their associated program would therefore revisit an established association.

The proposed contribution instead concerns the conditional relationship between an earlier wound response and subsequent activation of that program.

This distinction matters because two different explanations can accommodate weak downstream activation. Under a reduced-input explanation, refractory animals would mount a weaker early response, but a given response intensity would retain the same relationship to later RIC/ECM activation as in competent animals. Under the coupling hypothesis, comparable early responses would be followed by different downstream activation according to developmental stage. The proposed comparison therefore asks whether diminished early signaling is sufficient to account for reduced downstream activation, rather than treating lower expression at either time as evidence of defective coupling.

The prediction should be localized to the wound epidermis and ordered in time: an early post-amputation wound-response measure should precede the later epidermal transition being evaluated. A stage-dependent difference in this conditional relationship need not imply a particular numerical threshold or a weaker response slope; it could instead appear as lower downstream activation across comparable early-response intensities. The supplied evidence establishes neither the shape nor the magnitude of that relationship. Comparisons would also be informative only where the stages have overlapping early-response intensities, because an apparent conditional difference outside that overlap would depend on extrapolation.

Even a confirmed conditional difference would not, by itself, separate defective signal transmission from an intrinsic inability to enter the RIC/ECM state. Both could produce a weak downstream response despite a substantial early injury response. The conditional hypothesis thus provides an initial discrimination against a simple shortage of early input, not a complete mechanistic diagnosis. Nor would a difference between stage groups establish prediction of individual regeneration outcomes: competence varies between animals, so stage cannot substitute for an observed regenerative phenotype.

The available evidence supports posing this dependency as a bounded hypothesis, but does not establish its novelty, effect size, predictive advantage over stage-only or known-program baselines, or independent biological confirmation. Its value is to change the explanatory target from whether a regenerative program is reduced to whether its relationship with preceding injury responses changes during the refractory period. That shift makes the developmental switch a question about conditional responsiveness, rather than another description of an already recognized regenerative cell state.

4 What the transcriptomic comparison can identify

The transcriptomic comparison can constrain the proposed injury-to-regeneration dependency only by separating stage-level temporal responses from relationships measured across biological replicates. For Xenopus laevis, the proposed starting resource is GEO GSE245320, with bulk RNA-sequencing series GSE245317 and GSE245318, single-cell series GSE245312, and spatial series GSE245313. The candidate bulk comparison specified in the research brief comprises stages 40 and 46, sampled at 0, 0.5, and 6 hours and 3 days after amputation.

These are candidate design coordinates, not an inspected sample inventory: accession-level confirmation of sample availability, expression units, replication, and specimen relationships is necessary before assigning estimable contrasts.

The biological replicate must be defined before expression measurements are combined. Metadata would need to establish whether each library represents an individual tadpole or pooled material, whether libraries share contributing specimens, and how developmental stages and sampling times were allocated across batches. If independently collected pools are the available replicates, inference must remain at the pool level rather than treating their constituent animals as separately observed units. Likewise, technical libraries or cells derived from the same biological material cannot supply independent replication; if stage and batch are inseparable, their contributions cannot be distinguished by adding a batch term to a model.

With adequate independent replication and a separable batch structure, the bulk series could estimate responses over time within each stage and test whether those responses differ between stages. Comparisons against the corresponding stage-specific 0-hour samples would distinguish changes following amputation from expression differences already present at the candidate baseline. An early wound-response program and a subsequent RIC/ECM-associated program could then be evaluated as prespecified readouts rather than selected solely because they separate the stages. A stage-dependent temporal pattern would identify a difference in response trajectory, but would not by itself establish that the early response controls the later program.

The stronger conditional hypothesis requires information that a sequence of stage means does not provide. If early and late samples come from different animals or unrelated pools, an early measurement cannot be assigned to a particular later response merely because both samples share a developmental stage. Such sampling could show that one stage has an early response followed by weaker later activation, yet it would not estimate whether biological units with comparable early responses differ in their subsequent activation. Testing that dependency requires a defensible relationship between early and late measurements at the unit of inference, together with sufficient variation and overlap in early-response intensity across stages.

The single-cell and spatial series have complementary, narrower roles in this comparison. The supplied description identifies GSE245312 as predominantly characterizing regenerating animals, rather than providing matched single-cell snapshots across competent and refractory conditions; its intended role is therefore to inform the cellular interpretation of candidate programs, not to supply an assumed stage-matched test. GSE245313 could contribute anatomical localization where its verified sampling permits, without automatically establishing temporal continuity between early signaling and later activation.

Because these series belong to the same study, agreement across modalities would strengthen interpretation within that study but would not constitute independent biological confirmation.

Any conditional model should be evaluated only after simpler stage-only, RIC/ECM-only, and known-signaling-module baselines have been specified. Comparisons should quantify uncertainty at the biological-replicate level, account for multiple testing where applicable, and keep related measurements together when biological units are held out for evaluation. An interaction that is unstable across replicates or inseparable from stage should not be interpreted as improved prediction. Moreover, without verified animal-level regeneration outcomes, performance in distinguishing developmental stages cannot be described as performance in predicting which tadpoles regenerate.

The resulting distinction narrows the evidentiary contribution of the transcriptomic resource: stage-resolved trajectories could motivate an injury-to-regeneration coupling hypothesis, whereas a condi-

tional predictor requires linked biological-unit information beyond those trajectories. The decisive analytical question is therefore not simply whether early and late programs differ between stages, but whether the sampling structure permits their dependency to be estimated independently of stage membership. That distinction changes the central argument from one about a recognizable refractory-stage expression pattern to one about a specifically testable relationship between injury response and subsequent regenerative activation.

5 Separating cell-state coupling from alternative explanations

The coupling hypothesis is discriminating only if weaker later epidermal activation cannot be explained by fewer contributing cells, altered tissue composition, metabolic state, or generalized wound healing. An apparent relationship between early wound-response intensity and later RIC/ECM expression would not, by itself, distinguish these alternatives from defective intracellular or intercellular signaling. The relevant contrast is therefore not simply whether refractory-stage tissue expresses less of the later program, but whether comparable wound-epidermal populations activate that program less strongly at comparable early-response intensity.

Cell abundance and cell state require separate interpretations. Lower bulk RIC/ECM expression could reflect weaker activation within contributing cells, fewer cells entering the transient state, or a smaller epidermal contribution to the sampled tissue. These possibilities imply different mechanisms: an abundance difference could be a genuine developmental constraint without demonstrating a defect in how individual cells translate injury signals into regenerative activation. Proposed composition controls should consequently distinguish the representation of the basal epidermal population from the representation and activation of RIC-like states.

Because entry into the transient state is itself a candidate biological response, adjusting for RIC-like abundance alone could remove the phenomenon requiring explanation; abundance and within-state activation should instead be reported as distinct outcomes.

The single-cell series GSE245312 could provide a reference for constructing cell-identity and cell-state marker panels, rather than serving as a direct test of the refractory-stage transition. The supplied evidence describes it as predominantly characterizing regenerating animals, not as matched single-cell observations across competent and refractory conditions. A reference-derived annotation would therefore need to leave open whether refractory cells occupy states poorly represented in that reference.

Cells from one animal or library must not be treated as independent biological replicates, and agreement among the bulk, single-cell, and spatial series would provide complementary resolution within the same study rather than independent biological confirmation.

Spatial localization offers a further proposed test of whether a bulk association belongs to the wound epidermis. Subject to the specimens and measurements actually available in GSE245313, the analysis should ask whether the candidate later program localizes to the relevant epidermal region and whether its apparent reduction persists when local tissue composition is considered. A signal explained by different proportions of epidermis and adjacent tissues would favor a mixture explanation over reduced activation within comparable epidermal populations. Conversely, a

localized reduction at similar estimated epidermal abundance would make that explanation less sufficient.

Spatial proximity would still not establish that early and late signals occur in the same cells or identify the direction of communication between them.

Generalized wound healing is a separate alternative to regeneration-specific coupling. The proposed control is to compare the candidate later response with a separately specified general wound-response program, using comparable sampling times and anatomical regions wherever possible. Developmentally matched unamputated controls, if available or collected prospectively, would help distinguish injury-associated changes from differences already present before amputation. If the early-to-late relationship is equally evident for broad healing responses, or disappears when those responses are considered, it would not specifically implicate the regenerative transition.

The more informative pattern would be selective attenuation of later RIC/ECM activation despite a comparable measured early response, rather than a uniform reduction across injury-associated programs.

Metabolic gating could also produce the proposed conditional pattern without a primary defect in signal coupling. Work in Xenopus tropicalis implicates nutritional state in stage-dependent competence, but the supplied evidence also warns that cross-species stage behavior is context-dependent and not fully consistent; it does not establish the same explanation in X. laevis. For the latter species, nutritional or metabolic measurements would therefore be proposed explanatory variables, not assumed causes. A prospective comparison would ask whether the stage-dependent relationship between early response and later epidermal activation persists among animals with comparable measured metabolic states.

Transcriptomic associations alone would not determine whether metabolism limits the transition, accompanies it, or reflects tissue composition.

These alternatives sharpen rather than merely qualify the central hypothesis. If composition, abundance, or generalized healing accounts for the apparent association, the explanatory target shifts away from a distinct injury-to-regeneration coupling defect. If a selective reduction in later activation remains within comparable wound-epidermal populations at comparable early-response intensity, the hypothesis becomes more specific, although an intrinsic inability to enter the transient state would remain an alternative. The decisive explanatory quantity is thus the conditional activation of the epidermal program, not its bulk abundance or its ability to distinguish developmental stages.

6 Falsification, recovery, and the threshold for a mechanism

A paired rescue-and-block framework would test whether refractory-stage Xenopus laevis retain a regenerative epidermal transition that early injury signals fail to engage. The proposed comparison is between restoring downstream activation in stage-46 animals and selectively interrupting injury-to-transition coupling in stage-40 animals. Its value lies in testing the connection between injury and regeneration, rather than merely showing that a familiar regenerative program can be increased or decreased. The available evidence does not identify a validated intervention target, effective timing,

or experimental outcome, so this remains a causal framework rather than an executable protocol.

The framework would distinguish two mechanistic hypotheses. Under coupling failure, the refractory wound epidermis would retain the capacity to enter the transient regenerative state, but an early wound response would be insufficiently translated into that transition. Under intrinsic transition failure, activating the proposed connecting step would still fail to establish the later epidermal program. A minimal causal model would therefore separate early injury response, transient epidermal activation, extracellular-matrix remodeling, and tissue regrowth, allowing each proposed intervention to be interpreted against intermediate measurements rather than morphology alone.

The rescue arm would ask whether activating a step downstream of the early injury response restores the later epidermal program and regeneration in stage-46 animals. Rescue without a corresponding increase in the measured early response would be more consistent with a defect in signal transmission than would rescue produced simply by amplifying injury signaling. However, successful bypass would establish only that some downstream regenerative capacity remains accessible; it would not, by itself, identify the endogenous coupling defect.

Conversely, increasing a general injury signal alone would be ambiguous: improvement could reflect compensation for weak input, whereas failure could reflect either defective coupling or an inaccessible downstream transition.

The complementary block arm would ask whether disrupting the candidate connecting step in stage-40 animals suppresses later epidermal activation and regrowth while preserving the early wound response. That pattern would support a requirement for the proposed connection, provided the intervention did not instead compromise tissue viability or general wound healing. Directly suppressing the entire regenerative epidermal program would be less discriminating, because both coupling failure and intrinsic transition failure could predict loss of regeneration after such a block.

The informative pair would therefore restore access to the transition in refractory animals and interrupt that access in competent animals, rather than merely manipulate a shared downstream marker.

Target engagement would have to be verified independently of the outcomes used to claim rescue or blockade. An eventual design would measure the early wound response before the proposed regenerative transition, followed by wound-epidermal RIC activation, matrix-remodeling readouts, and subsequent regrowth. The later measurements should distinguish activation within the relevant epidermal population from changes in its abundance. Timing would need to be specified before testing, so that an intervention intended to alter coupling could be distinguished from one acting only after the transition or directly on subsequent tissue growth.

Several outcome patterns would constrain the hypothesis differently. Failure to restore the epidermal transition despite verified engagement of the proposed connecting step would count against that specific coupling-rescue model and would favor an intrinsic transition defect or another downstream restriction. Restoration of epidermal activation without regrowth would show that this transition alone is insufficient to explain regenerative failure. Likewise, loss of regrowth after blockade without loss of the predicted epidermal response would not support the proposed injury-to-epidermis mechanism. Failure to engage the target, by contrast, would leave the biological hypothesis untested rather than falsified.

Interpretation would require stage-matched sham, injury, and intervention controls, together with assessments of toxicity and developmental progression. Stage-40 and stage-46 animals should be evaluated within X. laevis; evidence from X. tropicalis would not establish the same stage-dependent response. A quantitative primary endpoint, such as blinded morphological regeneration scoring at a prespecified post-amputation time, would need to be paired with the intermediate molecular readouts.

Sample-size assumptions and uncertainty estimates should be defined at the biological-replicate level, without treating multiple cells or measurements from one specimen as independent animals; the supplied evidence does not support a numerical power calculation.

The threshold for a mechanism would be concordant rescue and blockade, verified target engagement, the predicted ordering of early and later responses, and confirmation in independently generated biological material. Prior-art assessment would also have to establish that the tested dependencynot simply RIC activation or matrix remodelinghad not already been demonstrated. The evidence therefore changes the central argument from a claim that the refractory period is caused by failed coupling to a sharper requirement: restoring downstream access must recover regeneration despite preserved early injury signaling, and selectively removing that access must reproduce failure in competent animals.

Those counterfactual outcomes would determine whether coupling explains the developmental switch or merely accompanies it.

7 Findings and hypotheses

7.1 C190 – Established

The investigation targets a mechanistically discriminating explanation of developmental loss and recovery of tail-regeneration competence in Xenopus laevis, including variation between animals. This establishes the scope of the research question, not an experimentally verified explanation of the developmental switch. The project explicitly excludes stage classification alone as evidence of animal-level prediction. *Evidence:* project:01de54c9-9593-413a-9577-1a952d28da70

7.2 C492 – Unresolved

RIC formation, basal-epidermal origin, and extracellular-matrix remodeling are identified as prior art in the project brief; their underlying literature has not been independently substantiated in the supplied selection. The brief attributes these findings to a 2024 Genome Biology study, but the selection supplies neither a complete citation nor source-level results. They may define the intended novelty boundary, but cannot support a detailed literature review here. *Evidence:* project:01de54c9-9593-413a-9577-1a952d28da70, strategy:b1010c01-7a62-4b61-912a-4560b0000001

7.3 C944 – Limitation

No complete research artifact, objective biological result, or reproducible computational result is available in the supplied selection. The indexed records repeatedly state that no complete artifact was saved and no objective result was established. This supports an evidence-availability limitation, not rejection of any biological hypothesis and not a claim about all potentially existing research. *Evidence:* session:6faba8dd-2aad-4cd1-9d4b-80bfef242140, session:084edf65-b9bb-45f4-9eec-8d55d0fc151c, session:757e88e6-270c-47f4-b0df-0c2acecc2fee, session:ef8d004e-68ff-4e2d-ab23-bb99a4f77df6, session:56860db5-f69c-496f-b386-0ceb4db56f4a, session:3ac1b431-bdde-4956-a2b0-05b0806b19f9

7.4 C516 – Conjecture

A candidate coupling-failure hypothesis predicts weaker subsequent epidermal RIC and extracellular-matrix program activation at stage 46 than at stage 40, conditional on comparable early wound-response intensity. This is a directional formulation of the temporal-coupling candidate already proposed in the brief and strategy descriptions. It distinguishes diminished downstream responsiveness from reduced upstream injury signaling, but no measured conditional effect or specific molecular mediator is supplied. *Evidence:* project:01de54c9-9593-413a-9577-1a952d28da70, strategy:b1010c01-7a62-4b61-912a-4560b0000002, strategy:b1010c01-7a62-4b61-912a-4560b0000004

7.5 C556 – Unresolved

The actual sample structure and inferential suitability of GSE245320 and its listed component series remain unverified in the supplied evidence. The brief names bulk series GSE245317 and GSE245318, single-cell series GSE245312, spatial series GSE245313, and candidate stages and sampling times. The selection contains no retrieved sample manifest establishing replicate counts, pooling, batches, available files, phenotype labels, or coverage of particular stagetime combinations. *Evidence:* project:01de54c9-9593-413a-9577-1a952d28da70, strategy:b1010c01-7a62-4b61-912a-4560b0000001, session:6faba8dd-2aad-4cd1-9d4b-80bfef242140

7.6 C331 – Limitation

Neither an animal-level early-to-late coupling estimate nor an animal-level regeneration predictor can be inferred from the supplied records. Individual outcome labels and links between early and later specimens have not been verified. The brief also cautions that the single-cell evidence predominantly concerns regenerating animals and that the listed modalities share a study. Stage means, individual cells, and multiple modalities therefore cannot be assumed to supply paired animals or independent replication. *Evidence:* project:01de54c9-9593-413a-9577-1a952d28da70, strategy:b1010c01-7a62-4b61-912a-4560b0000002, strategy:b1010c01-7a62-4b61-912a-4560b0000003

7.7 C723 – Unresolved

The proposed temporal interaction has no reported numerical effect, uncertainty estimate, or demonstrated advantage over stage-only, RIC/ECM-only, and known-signaling baselines. The strategy descriptions propose these comparisons, but the indexed outcome records contain no fitted models or performance estimates. Neither biological-mechanism success nor the lower-tier computational result can be claimed. *Evidence:* strategy:b1010c01-7a62-4b61-912a-4560b0000002, strategy:b1010c01-7a62-4b61-912a-4560b0000005, session:757e88e6-270c-47f4-b0df-0c2acecc2fee

7.8 C373 – Unresolved

Tissue composition, metabolic gating, generalized healing, batch structure, and developmental differences remain unresolved alternatives to a cell-state coupling mechanism. These alternatives are explicitly included in the hypothesis space and proposed controls. No spatial, compositional, metabolic, or batch-adjusted result is supplied that separates them from the coupling hypothesis. *Evidence:* project:01de54c9-9593-413a-9577-1a952d28da70, strategy:b1010c01-7a62-4b61-912a-4560b0000002, strategy:b1010c01-7a62-4b61-912a-4560b0000003

7.9 C232 – Unresolved

The novelty of a stage-dependent injury-to-RIC/ECM coupling rule has not been established. The selection describes intended prior-art and adversarial novelty searches but supplies no completed cited ledger. A hypothesis suggested in the project brief is not thereby a previously unreported finding. *Evidence:* strategy:b1010c01-7a62-4b61-912a-4560b0000001, strategy:b1010c01-7a62-4b61-912a-4560b0000005, session:084edf65-b9bb-45f4-9eec-8d55d0fc151c

7.10 C240 – Unresolved

Reciprocal rescue and blocking interventions are proposed as a falsification framework, but no molecular target, executable protocol, or independent biological confirmation is established. The saved descriptions propose a stage-46 rescue and corresponding stage-40 block with timing, injury and sham controls, toxicity assessment, and a blinded regeneration endpoint. They do not document a selected perturbation, conducted experiment, effect size, or independent validation cohort. *Evidence:* strategy:b1010c01-7a62-4b61-912a-4560b0000004, strategy:b1010c01-7a62-4b61-912a-4560b0000005, session:3ac1b431-bdde-4956-a2b0-05b0806b19f9

7.11 C279 – Limitation

The proposed stage-40 versus stage-46 comparison alone does not establish an explanation for the later recovery of regeneration competence. Later recovery is part of the motivating question, whereas the described observational contrast centers on competent and refractory stages. The supplied selection contains no verified recovery-stage observations that test whether the proposed

coupling returns. *Evidence:* project:01de54c9-9593-413a-9577-1a952d28da70, strategy:b1010c01-7a62-4b61-912a-4560b0000002

8 Limitations

- The supplied selection supports a hypothesis-focused article plan, not an empirical results paper. It contains no complete research artifact with expression analyses, numerical effects, uncertainty estimates, or independent validation.
- The literature background and accession descriptions come from project context and proposed strategies rather than retrieved publications and verified sample metadata. No exhaustive literature review, novelty determination, or accession-level analysis can be represented as completed.
- The candidate lacks an identified molecular mediator and an executable perturbation protocol. Even a future conditional transcriptomic association would require additional evidence to establish causality and explain later recovery.
- Cross-species nutritional observations mentioned in the brief cannot be transferred from Xenopus tropicalis to Xenopus laevis without species-specific evidence.

9 Open obligations

- Retrieve the underlying publications and complete a cited prior-art ledger through a documented investigation cutoff, including explicit searches for the proposed conditional temporal relationship.
- Verify all relevant GEO samples, files, developmental stages, sampling units, biological replicates, pooling, batches, and phenotype labels before defining the estimable analysis.
- Establish whether early and later measurements can be linked at an appropriate biological level. If specimens are unpaired, restrict inference to supported population-level contrasts rather than constructing artificial animal trajectories.
- Specify early-response and later epidermal activation measurements, their timing, the primary conditional contrast, and uncertainty estimation before fitting the candidate interaction.
- Fit stage-only, RIC/ECM-only, and known-signaling baselines; assess composition, batch, stage, and metabolic alternatives with prespecified negative controls, multiplicity safeguards, and biological-replicate-aware validation.
- Select and justify a molecular perturbation, reciprocal rescue and blocking conditions, timing controls, sham and injury controls, toxicity assessment, blinded regeneration endpoint, and sample-size assumptions. Define counterfactual outcomes that distinguish coupling failure from intrinsic transition failure.

- Obtain independently generated biological evidence on previously unseen specimens or conditions, with individual regeneration outcomes where animal-level prediction is claimed.
- Test a later competent developmental condition to determine whether the proposed coupling recovers alongside regenerative ability.

10 Conclusions

Stage-dependent injury-to-regeneration coupling is a testable candidate explanation, not an established or demonstrated novel mechanism. The supplied evidence does not support a numerical association, improved predictor, or causal result.

The decisive distinction is between reduced upstream injury signaling and reduced downstream regenerative activation conditional on that signaling. Resolving it requires verified biological replication, controls for composition and developmental confounding, and independent evidence that addresses both refractory-stage failure and later recovery.

Authorship note

AI-assisted synthesis from saved research; not independently peer reviewed.